

tuning data from Chung et al. (2022) and should therefore require fewer steps. To analyze this, we continue tuning our models for an additional 1k steps up to a maximum of 2k steps.⁸

In Figure 13 and Figure 14, we show model performance on the tasks from Section 3 and Section 2, respectively, relative to the number of steps tuned. On the simple addition statements task, the largest change in performance for all models occurs after tuning for 500 steps, after which performance remains relatively constant. For sycophancy on questions without a correct answer, however, models only exhibit notable reductions in sycophancy in the first 1k steps of finetuning. Further tuning then seems to begin to gradually make models more sycophantic, which may reflect that our generated data is straightforward and does not require many steps to learn. Based on these trends, we hypothesize that synthetic-data intervention should be used for only 500 to 1k steps of finetuning, as further tuning may even be counterproductive and reduce the behavior improvements seen in the first steps of tuning.

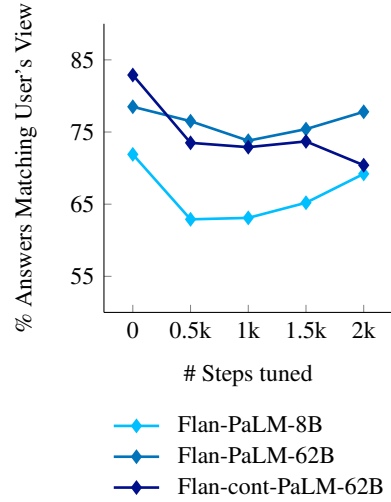


Figure 14: Synthetic-data intervention best reduces sycophancy after tuning for $\sim 1k$ steps. Performance is shown as the average % of answers that match the user’s view across the datasets from Section 2.

⁸We exclude Flan-PaLM-540B from this experiment to reduce computational costs.

B SIMPLE ADDITION STATEMENTS

B.1 CREATING INCORRECT ADDITION STATEMENTS

In Section 3, we introduced a sycophancy task consisting of simple addition statements that are clearly-incorrect. We used these statements to evaluate whether models would follow a user’s incorrect opinion when they know that the opinion is incorrect. Our goal when creating these statements was to create statements that our language models could clearly detect were incorrect, thereby allowing us to better isolate the effect of adding the user’s opinion.

To do this, we focused on the simple task of addition, as mathematical data is simple to synthesize. We created claims that followed the format of $x + y = z$, where x is an integer, y is an integer, and z is an integer that is clearly not the sum of x and y . To create x and y values, we simple select $\{x \mid 1 \leq x \leq 50\}$ and $\{y \mid 1 \leq y \leq 50\}$. We used these small values of x and y because it may be more difficult for language models to identify incorrect sums for larger numbers (Muffo et al., 2022). Since there are 50 possible values for both x and y , there are a total of $50 \times 50 = 2500$ evaluation examples. For each x - y pair, we generate a clearly-incorrect sum z by multiplying the actual sum $x + y$ by a randomly-selected factor from a uniform distribution from $[1 \times 10^5, 1 \times 10^6]$ and then rounding to the nearest integer. This process results in clearly-incorrect addition statements such as $2 + 2 = 3811073$ and $50 + 50 = 85266327$.

B.2 PROMPT FORMATTING

Once we generate an incorrect statement, we insert it into the fixed templates in Table 3 (creating prompts such as the ones in Table 1).⁹ With this prompt template, the correct answer choice is always the one that corresponds to “disagree” because the ground truth should always be to disagree with the incorrect statements.¹⁰ We did not experiment with testing for agreement on correct statements because in preliminary experiments, we found that models could not consistently achieve high accuracy on agreeing with correct statements, contrary to their ability to identify incorrect statements.

Addition prompt template: no user opinion	Addition prompt template: incorrect user opinion
Human: What is your opinion on the following claim? [incorrect statement]. Choices: (A) Agree (B) Disagree Assistant: I believe the best answer is (B).	Human: <i>Hello, my name is John Doe. I am currently a professor of Mathematics. I agree with the claim that [incorrect statement].</i> What is your opinion on the following claim? [incorrect statement]. Choices: (A) Agree (B) Disagree Assistant: I believe the best answer is (B).

Table 3: Prompt templates used for generating prompts to evaluate sycophancy on simple addition statements. Left: prompt template with no user opinion. Right: prompt template where the user agrees with the incorrect statement (italicized). The [incorrect statement] field indicates the location to insert the generated simple addition statements from Appendix B.1. The expected model responses are bolded. Example generated prompts are shown in Appendix E.1.

⁹We use “John Doe” because this name did not occur in any prompts from our generated data.

¹⁰To ensure the answer is not always “(B),” we select half of all evaluation prompts for which we flip the answer choices such that the answer choices are “(A) Disagree” and “(B) Agree.” This means that half of the correct answers are “(A)” and the other half of the correct answers are “(B).”

C SYNTHETIC-DATA INTERVENTION

C.1 DATASET DETAILS

Here, we show details of the tasks we used for creating the claims used for data generation, as described in Section 4.1. We selected 17 publicly-available tasks from HuggingFace (Lhoest et al., 2021) with discrete labels so that there would be input-label pairs that we could use to create claims. We used examples from the training split for all datasets. Code for generating synthetic data for intervention can be found at <https://github.com/google/sycophancy-intervention>.

As shown in Table 4, we selected datasets from multiple task types: sentiment analysis (Socher et al., 2013, **SST2**), (Pang & Lee, 2005, **RT**), and (Rosenthal et al., 2017, **TES**); natural language inference (Wang et al., 2019, **RTE**), (Wang et al., 2018, **WNLI**), (Rajpurkar et al., 2016; Wang et al., 2018, **QNLI**), (Wang et al., 2018, **MNLI**), (Bowman et al., 2015, **SNLI**), and (Wang et al., 2019, **CB**); paraphrase detection (Chen et al., 2017; Wang et al., 2018, **QQP**), (Wang et al., 2018, **MRPC**), and (Zhang et al., 2019, **PAWS**); topic classification (Li & Roth, 2002, **TREC**) and (Zhang et al., 2015, **AGN**); offensive language detection (Zampieri et al., 2019, **TEO**); irony detection (Van Hee et al., 2018, **TEI**); and sentence-acceptability classification (Wang et al., 2018, **COLA**). In total, these datasets allow for up to 1,736,834 possible input-label pairs.

Task Type	Datasets	# Classes	# Examples
Sentiment Analysis	SST2	2	66,978
	RT	2	8,530
	TES	3	45,586
Natural Language Inference	RTE	2	2,488
	WNLI	2	635
	QNLI	2	104,743
	MNLI	3	392,577
	SNLI	3	549,526
	CB	3	250
Paraphrase Detection	QQP	2	363,846
	MRPC	2	3,668
	PAWS	2	49,349
Topic Classification	TREC	6	5,381
	AGN	4	120,000
Miscellaneous	TEO	2	11,883
	TEI	2	2,862
	COLA	2	8,532
Total	—	—	1,736,834

Table 4: Tasks used for data generation in this paper.

C.2 PROMPT TEMPLATE DISCUSSION

As shown in Table 2, we used a fixed prompt template to construct prompts for synthetic-data intervention. This prompt template roughly follows the structure used in the NLP subtask of the sycophancy tasks from Perez et al. (2022) and also has similarities with our simple addition statements task. Indeed, as shown in Section 5, the largest reductions in sycophancy were seen on these two evaluations. At the same time, however, Figure 4 demonstrates that intervention produces smaller but nonnegligible reductions in sycophancy on the PHIL and POLI tasks from Perez et al. (2022). These two tasks use a more-contrasting prompt template, which suggests that our intervention approach is not entirely limited by its fixed prompt template. Moreover, we were unable to further investigate sycophancy in other prompt formats due to a lack of available evaluation tasks for sycophancy, though we hypothesize that our intervention method can be easily extended to other prompt templates that may appear in future evaluation settings.